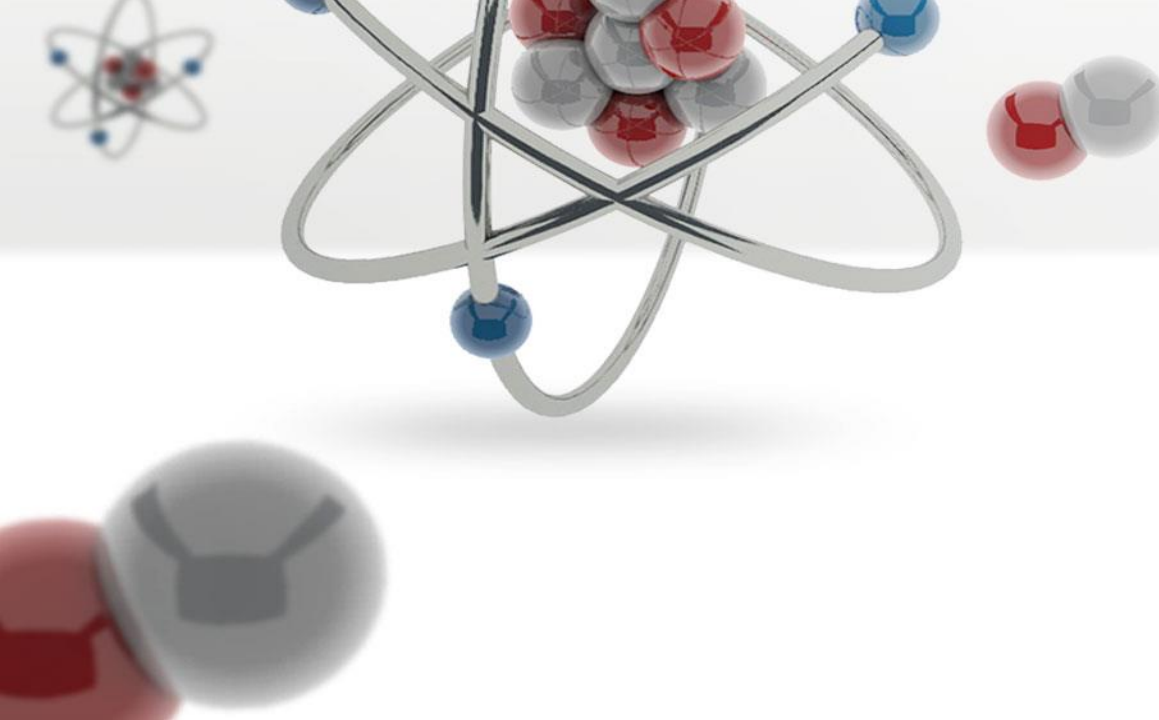




Computational Methods for Quantum Many-Body Systems (CMQMB) - from **artificial atoms** to **high-temperature superconductors**

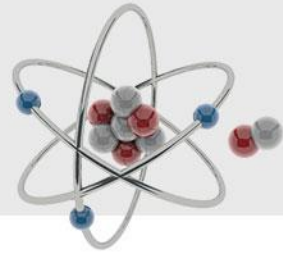
Lecture 4 – From single- to many-body Green functions



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Prof. Dr. Thomas Schäfer

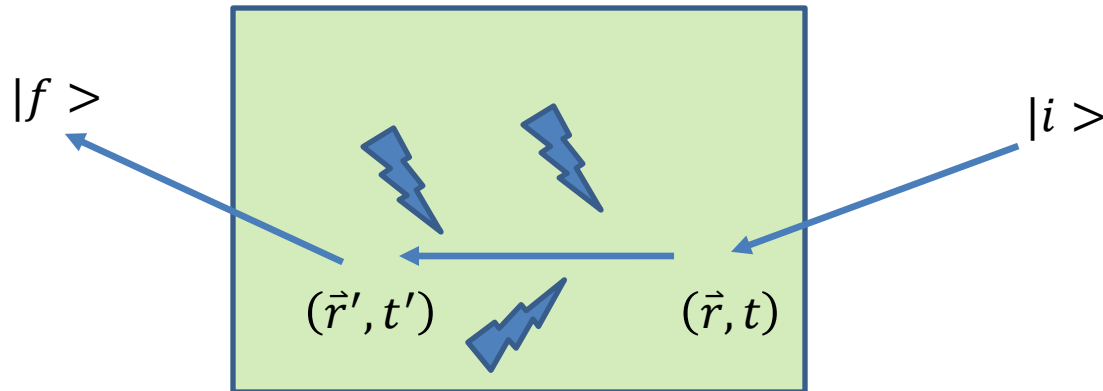
Università degli studi di Trieste (UNITS), Winter Semester 2025



Towards interacting systems...

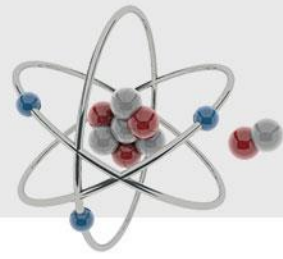
How can we obtain information from an interacting system?

→ Scattering experiments (theory)



Three ingredients necessary:

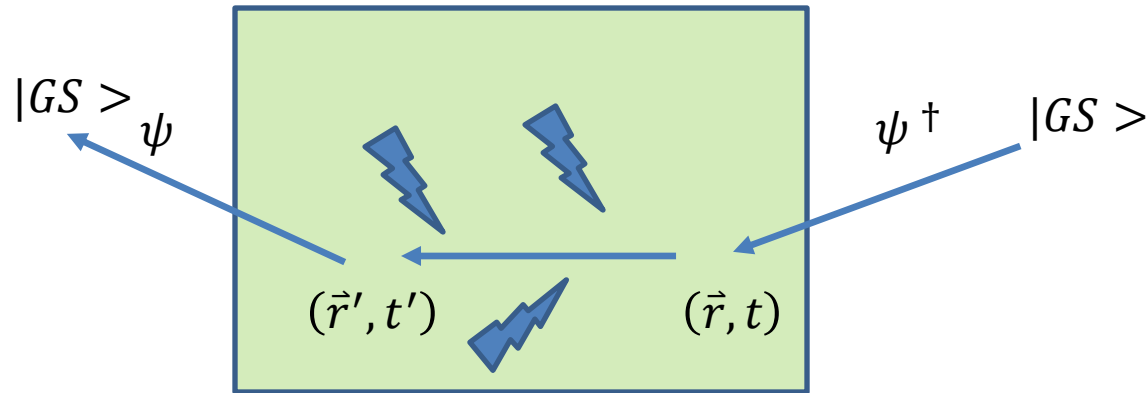
- 1) Many-body treatment → Second quantization ✓
- 2) Time evolution → Pictures of time evolution
- 3) How does the system respond? → Linear response theory



Towards interacting systems...

How can we obtain information from an interacting system?

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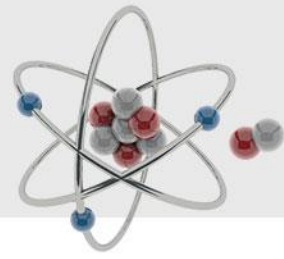


→ Green functions

Three ingredients necessary:

- 1) Many-body treatment → Second quantization ✓
- 2) Time evolution → Pictures of time evolution ✓
- 3) How does the system respond? → Linear response theory ✓

Lecture 4



Content and goals

- What is a Green function? Example of the one-particle Schrödinger equation
- Many-body Green functions
 - Definition
 - Translational invariance, basis change
 - Green function of free electrons
- Lehmann representation [and spectral function]